INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

SALTLAKE

(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – I, Second Semester Examination, February – 2026

Subject Code: BCACC202

Subject Name: Object Oriented Programming

Full Marks: 30 | **Set No:1** | Duration: 1 Hour

Part – A

Answer any 5 questions out of 6

Each question carries 2 marks (2 × 5)

1. What are the key principles of Object-Oriented Programming (OOP)? **2**
2. Differentiate between structure and class in C++? **2**
3. Define constructor and destructor with examples. **2**
4. How member function is differed from the friend function? **2**
5. What is scope resolution operator? **2**
6. What are the errors in the following program:- **2**

```
#include iostream
using namespace std

void main() {
   int a = 10
   if (a = 5)
      cout << 'Value is 5';
}
```

Part – B

Answer any 2 questions out of 3

Each question carries 5 marks (5 × 2)

7. Difference between object oriented programming and procedure oriented programming. **5**
8. Describe data types in C++ in Details. **5**

9. Develop a C++ program to find out the factorial of a given number, n with the help of object and class concept. 5

Part – C

Answer any 1 question out of 2
Each question carries 10 marks (10 × 1)

10. Write a C++ program to create a class **Student** with the following members: 10
    - Data members: Roll No, Name, Marks of 3 subjects
    - Member functions:
        - Input student details
        - Calculate total and average
        - Display all details

    Create objects for 2 students and display their results.

11. Develop a C++ program to store and display book information (title, ISBN, price etc.) with Using array of object concept. 10

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